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Q) In JAVA every object has Hashcode, find out what is Hashcode and why it is being generated in JAVA.

Every object in Java automatically gets a number attached to it, called a hash code. Think of it like a "locker number" for that object.

⇒ `Object obj = new Object();`

`System.out.println(obj.hashCode());` //print some number

Why does it exist?

Imagine HashMap or HashSet as a big cupboard with many small lockers (buckets). When we store an object, Java doesn't want to check every locker one by one to find where it goes - that would be slow.

So instead -

- Java uses the hashcode to instantly decide which locker (bucket) the object should go into.
- This makes adding, finding and removing objects very fast, instead of checking everything one by one.

With hashcode it jumps straight to where the object should be.

In simple words we can say that Hashcode is said as a number given to every object, used mainly to quickly find/store objects in collections like HashMap and HashSet, instead of checking each one manually.